

SRI VASAVI ENGINEERING COLLEGE (Autonomous)

(Permanent Affiliation to JNTUK, Kakinada), PEDATADEPALLI, TADEPALLIGUDEM-534 101



Department of Computer Science and Engineering

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Department of Computer Science and Technology

Course Outcomes of B.Tech(CSE) & B.Tech(CST) -V23 Regulation

Semester	Course Code & Course Name	Course Outcomes
I	V23CMT01 Basic Civil & Mechanical Engineering	After Successful completion of the Course, the student will be able to: Part - A: CO1: Apply knowledge of fundamental laws. to analyze KCL, KVL, series & parallel circuits, ac fundamentals, average, RMS values. (K3) CO2: Understand the construction and operation of AC and DC Electrical machines, measuring instruments (PMMC, MI). (K2) CO3: Understand the various sources of energy, types of power plants and layout representation of electrical power systems, electricity bill calculations and safety measures. (K2) Part - B: CO4: Understand the semiconductor devices –Characteristics of diodes, transistors. (K2) CO5: Understand the basic electronic circuits - Rectifiers and power supplies, Amplifiers & Instrumentation. (K2) CO6: Apply knowledge of Number Systems, Logic gates, Boolean algebra, Truth Tables to solve Simple combinational circuit. (K3)
I	V23CHT01 Applied Chemistry	After Successful completion of the Course, the student will be able to: CO1: Solution for the particle in the one dimensional box and applications of MO theory. (K3) CO2: Classify the advanced materials and its applications in Technology. (K3) CO3: Associate concepts of Electro chemical sensors and designing Electro chemical energy storages . (K3) CO4: Summarize the importance of polymers and conducting polymers. (K3) CO5: Compile the Advance spectroscopy and its applications. (K3)
I	V23MAT01 Linear Algebra & Calculus	After Successful completion of the Course, the student will be able to: CO1: Apply matrix algebra techniques to solve engineering problems. (K3) CO2: Use Eigen values and Eigen vectors concept to find nature of quadratic form, inverse and powers of matrix. (K3) CO3: Apply mean value theorems to real life problems. (K3) CO4: Use the concepts of functions of several variables which is useful in optimization. (K3) CO4: Apply double and triple integrals of functions of several variables in two dimensions using Cartesian and polar coordinates and in three dimensions using cylindrical and spherical coordinates. (K3)
I	V23CST01 Introduction to Programming	After Successful completion of the Course, the student will be able to: CO1: Understand basics of computers, the concept of algorithm and algorithmic thinking. (K2) CO2: Analyze a problem and develop an algorithm to solve it. (K3) CO3: Implement various algorithms using the C programming language. (K3) CO4: Understand more advanced features of C language. (K2) CO5: Develop problem-solving skills and the ability to debug and optimize the code. (K3)

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I	V23MET01 Engineering Graphics	After Successful completion of the Course, the student will be able to: CO1: Understand the principles of engineering drawing, including engineering curves, scales, orthographic and isometric projections. (K3) CO2: Draw and interpret orthographic projections of points, lines, planes and solids in front, top and side views. (K3) CO3: Understand and draw projection of solids in various positions in first quadrant. (K3) CO4: Explain principles behind development of surfaces. (K3) CO5: Prepare isometric and perspective sections of simple solids. (K3)
I	V23CSL02 IT Workshop Lab	After Successful completion of the Course, the student will be able to: CO1: Demonstrate Hardware troubleshooting (K3) CO2: Understand Hardware components and inter dependencies. (K2) CO3: Employ safeguard measures of computer systems from viruses/worms. (K3) CO4: Report Document/ Presentation preparation. (K2) CO5: Compute calculations using spreadsheets. (K3)
I	V23MEL01 Engineering Workshop Lab	After Successful completion of the Course, the student will be able to: CO1: Identify workshop tools and their operational capabilities. (K2) CO2: Practice on manufacturing of components using workshop trades including fitting, carpentry, foundry and welding. (K3) CO3: Apply fitting operations in various applications. (K3) CO4: Apply basic electrical engineering knowledge for House Wiring Practice. (K3)
I	V23CHL01 Applied Chemistry Lab	After Successful completion of the Course, the student will be able to: CO1: Determine the cell constant and conductance of solutions. (K4) CO2: Prepare advanced polymer Bakelite materials. (K3) CO3: Measure the strength of an acid present in secondary batteries. (K5) CO4: Analyse the IR spectra of some organic compounds. (K4) CO5: Calculate strength of acid in Pb-Acid battery. (K4)
I	V23CSL01 Computer Programming Lab	After Successful completion of the Course, the student will be able to: CO1: Read, understand and trace the execution of programs written in C language. (K3) CO2: Select the right control structure for solving the problem. (K3) CO3: Develop C programs which utilize memory efficiently using programming constructs like pointers. (K3) CO4: Develop, Debug and Execute programs to demonstrate the applications of arrays, functions, basic concepts of pointers in C. (K3)
I	V23SPT01 NSS/NCC/Scouts & Guides/Community Service	After Successful completion of the Course, the student will be able to: CO1: Understand the importance of discipline, character and service motto. (K2) CO2: Solve some societal issues by applying acquired knowledge, facts, and techniques. (K3) CO3: Examine human relationships by analyzing social problems. (K3) CO4: Determine to extend their help for the fellow beings and downtrodden people. (K4) CO5: Develop leadership skills and civic responsibilities. (K3)
II	V23MAT02 Differential Equations & Vector Calculus	After Successful completion of the Course, the student will be able to: CO1: Apply the techniques of ordinary differential equations of first order and first degree related to various engineering fields. (K3) CO2: Apply the techniques of linear differential equations of higher order related to various engineering fields. (K3) CO3: Solve the first order linear PDE and homogeneous PDE by appropriate methods. (K3) CO4: Interpret the physical meaning of gradient, curl and divergence. (K3) CO5: Find the line, surface and volume integrals and their relations using vector calculus techniques. (K3)
II	V23PHT01 Engineering Physics	After Successful completion of the Course, the student will be able to: CO1: Examine the intensity variation of light due to polarization, interference and diffraction. (K3) CO2: Describe the basics of crystals and their structures. (K2)

Semester	Course Code & Course Name	Course Outcomes
		CO3: Interpret the various types of polarization of dielectrics and classify the magnetic materials. (K3) CO4: Explain the basic concepts of Quantum Mechanics and apply it to one dimensional motion of particles. (K2) CO5: Examine the type of semiconductor using Hall effect. (K3)
II	V23ENT01 Communicative English	After Successful completion of the Course, the student will be able to: CO1: Identify the main idea of the text using skimming and scanning methods of reading and Parts of Speech. Frame Sentences and questions using appropriate Punctuation and Vocabulary. (K2) CO2: Interpret the poem and write summary, using appropriate vocabulary. Gain mastery over articles, prepositions and improve writing skills. (K3) CO3: Find the success formula after reading the text in detail to answer questions. Use appropriate tense and concord, and get acquainted with summarizing, note-making and paraphrasing. (K3) CO4: Employ reading skills to comprehend the given inspirational story. Find suitable vocabulary and format to draft Letters and Resume. (K3) CO5: Appraise the delivered lecture and text, recognize the contextual vocabulary, write Essays and prepare Presentations. (K4)
II	V23EET01 Basic Electrical and Electronics Engineering	After Successful completion of the Course, the student will be able to: CO1: Apply Knowledge of fundamental laws to analyze AC and DC Electrical circuits. (K3) CO2: Understand the construction and operation of electrical machines and measuring instruments. (K2) CO3: Understand the layout of various power plants, electricity bill and safety measures. (K2) CO4: Understand the working of various semiconductor devices. (K2) CO5: Understand the basic electronic circuits. (K2) CO6: Apply knowledge of digital electronics to solve various combinational circuits. (K3)
II	V23CST02 Data Structures	After Successful completion of the Course, the student will be able to: CO1: Explain the role of linear data structures in organizing and accessing data efficiently in algorithms. (K2) CO2: Design, implement, and apply linked lists for dynamic data storage, demonstrating understanding of memory allocation. (K3) CO3: Develop programs using stacks to handle recursive algorithms, manage program states, and solve related problems. (K3) CO4: Apply queue-based algorithms for efficient task scheduling and breadth-first traversal in graphs and distinguish between dequeues and priority queues, and apply them appropriately to solve data management challenges. (K3) CO5: Develop novel solutions to small scale programming challenges involving data structures such as Trees, hashing. (K3)
II	V23PHL01 Engineering Physics Lab	After Successful completion of the Course, the student will be able to: CO1: Operate optical instruments like travelling microscope and spectrometer. (K3) CO2: Estimate the wavelengths of different colours using diffraction grating. (K2) CO3: Plot the intensity of the magnetic field of circular coil carrying current with distance. (K3) CO4: Demonstrate the various experiments in the areas of optics, mechanics and Electronics. (K3) CO5: Calculate the band gap of a given semiconductor. (K3) CO6: Estimate Hall voltage and Hall coefficient of a given semiconductor using Hall effect. (K2)
II	V23ENL01 Communicative English Lab	After Successful completion of the Course, the student will be able to: CO1: Identify vowels, consonant sounds, neutralization and accent rules for better listening and speaking comprehension. (K1) CO2: Collect suitable expressions and vocabulary to participate in JAM and Role plays. (K1) CO3: Arrange your ideas to draft an email, Resume and SOP. (K1)

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		CO4: Organize your ideas and prepare to give presentations in a professional manner and Debate rationally and cogently. (K3) CO5: Show team spirit and communicative skills in group discussion and preparation to attend interviews. (K3)
II	V23EEL01 Electrical and Electronics Engineering Workshop Lab	After Successful completion of the Course, the student will be able to: CO1: Analyse the Electrical circuit design concept, power and power factor. (K4) CO2: Understand the operation and characteristics of Electrical Machines. (K2) CO3: Analyse various domestic loads of energy bill calculation. (K4) CO4: Understand the various characteristics of electronic devices. (K2) CO5: Development of the digital circuits for electronic device applications. (K3)
II	V23CSL03 Data Structures Lab	After Successful completion of the Course, the student will be able to: CO1: Explain the role of linear data structures in organizing and accessing data efficiently in algorithms. (K3) CO2: Design, implement, and apply linked lists for dynamic data storage, demonstrating Understanding of memory allocation. (K3) CO3: Develop programs using stacks to handle recursive algorithms, manage program states, and solve related problems. (K3) CO4: Apply queue-based algorithms for efficient task scheduling and breadth-first traversal in graphs and distinguish between dequeues and priority queues and apply them appropriately to solve data management challenges. (K3) CO5: Recognize scenarios where hashing is advantageous, and design hash-based solutions for specific problems. (K3)
II	V23SPT02 Health and wellness, Yoga and sports	After Successful completion of the Course, the student will be able to: CO1: Discuss the importance of yoga and sports for Physical fitness and sound health. (K2) CO2: Demonstrate an understanding of health-related fitness components. (K3) CO3: Compare and contrast various activities that help enhance their health. (K4) CO4: Assess current personal fitness levels. (K3) CO5: Develop Positive Personality. (K3)
III	V23MAT05 Discrete Mathematics & Graph Theory	After Successful completion of the Course, the student will be able to: CO1: Develop skills required for solving mathematical problems using mathematical logic. (K3) CO2: Demonstrate the set theory principles, relations and functions in real-time Situations. (K3) CO3: Apply the knowledge of combinatorics and recurrence relations in formulating and solving complex problems. (K3) CO4: Apply the graph theory principles and techniques in computer science-related Problems. (K3) CO5: Find shortest paths and minimal spanning trees using prim's and Kruskal's algorithms, BFS and DFS algorithms. (K3)
III	V23MBT51 Managerial Economics and Financial Analysis	After Successful completion of the Course, the student will be able to: CO1: Understand the basic concepts of managerial economics, demand, and elasticity of demand and methods of demand forecasting. (K2) CO2: Interpret production concept, least cost combinations and various costs concepts indecision making. (K3) CO3: Differentiate various Markets and Pricing methods a long with Business cycles (K2) CO4: Prepare financial statements and its analysis. (K3) CO5: Assess various investment project proposals with the help of Capital Budgeting techniques for decision making. (K3)
III	V23CST03 Digital Logic & Computer Organization	After Successful completion of the Course, the student will be able to: CO1: Explain different Data Representation and various Combinational Digital Logic Circuits. (K2) CO2: Explain various Sequential Digital Logic Circuits and basic Structure of Computers. (K2) CO3: Describe the basic concepts of Computer Arithmetic and Processor Organization. (K2)

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		CO4: Illustrate different types of Memory and organization of Memory.(K2) CO5: Demonstrate the ways of accessing various Interfacing devices with Processor.(K3)
III	V23CST04 Advanced Data Structures & Algorithm Analysis	After Successful completion of the Course, the student will be able to: CO1: Demonstrate asymptotic notation and non-linear data structures like AVL and B-Trees.(K3) CO2: Demonstrate graphs and divide and conquer technique.(K3) CO3: Use greedy approach to determine various problems.(K3) CO4: Apply Dynamic Programming and Backtracking techniques.(K3) CO5: Solve different problems using Branch and Bound ,NP Hard and NP Complete.(K3)
III	V23CST05 Object Oriented Programming Through Java	After Successful completion of the Course, the student will be able to: CO1: Demonstrate the object-oriented programming principles with Java programming environment. (K3) CO2: Demonstrate the concepts like classes, objects, argument passing mechanism, overloading and overriding. (K3) CO3: Illustrate the concepts of arrays, inheritance and interfaces. (K3) CO4: Demonstrate packages , java libraries, exception handling, java I/O and File concepts. (K3) CO5: Illustrate the concepts of string handling, multithreading and Java FX GUI. (K3)
III	V23CSL04 Advanced Data Structures and Algorithm Analysis Lab	After Successful completion of the Course, the student will be able to: CO1: Demonstrate programs on AVL Trees and Heap trees. (K3) CO2: Develop programs on Sorting algorithms and Graph traversal algorithms.(K3) CO3: Develop programs using Greedy and Dynamic programming technique. (K3) CO4: Develop programs using Backtracking and branch & Bound technique. (K3)
III	V23CSL05 Object Oriented Programming Through Java Lab	After Successful completion of the Course, the student will be able to: CO1: Construct programs to handle classes and objects. (K3) CO2: Develop programs that incorporate inheritance and interfaces. (K3) CO3: Construct programs on exception handling and File I/O. (K3) CO4: Develop programs using multithreading and Java FX. (K3)
III	V23CSSE01 Python Programming Lab	After Successful completion of the Course, the student will be able to: CO1: Illustrate basic concepts and control structures in python Programming. (K3) CO2: Demonstrate functions and packages. (K3) CO3: Construct python programs using structured data types. (K3) CO4: Develop programs on Files, Exception handling and OOPs Concepts. (K3) CO5: Construct programs for Data Analysis using Num Py and Pandas. (K3)
III	V23MET09 Design Thinking & Innovation	After Successful completion of the Course, the student will be able to: CO1: Define the concepts related to design thinking. (K1) CO2: Explain the fundamentals of Design Thinking and innovation.(K2) CO3: Apply the design thinking techniques for solving problems in various sectors.(K3) CO4: Analyse to work in a multidisciplinary environment. (K4) CO5: Evaluate the value of creativity. (K5)
III	V23ENT02 Professional Communication Skills - I	After Successful completion of the Course, the student will be able to: CO1: Use vocabulary in regular chores of life with accuracy, make meaningful sentences, and describe people and their traits vividly. (K3) CO2: Distinguish between places of pilgrimage and holiday spots; describe incidents, things and process; and frame questions, statements and expressions. (K4) CO3: Demonstrate their knowledge of idioms which are similar to those of native speakers while speaking and writing and use phrases clearly and precisely to articulate their views that compare and contrast indianisms with native expressions and avoid common errors. (K3) CO4: Employ the vocabulary of netizens with ease and walk through the letters and emails for effective official correspondence and infer the accurate meaning of the homophones that are often confusing. (K3)

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		CO5: Summarize their profile; introduce themselves as well as others by incorporating their accomplishments and Sketch stories and anecdotes in an interesting and engaging manner that arouses curiosity of the audience. (K5)
IV	V23MBT53 Universal Human Values	After Successful completion of the Course, the student will be able to: CO1: Understand the fundamentals of Natural Acceptance, Happiness and Prosperity. (K2) CO2: Identify one's self, and one's surroundings(family, society, nature). (K2) CO3: Apply what they have learnt to their own self in different day-to-day settings in real life. (K3) CO4: Explain human values with human relationship and human society. (K2) CO5: Identify the need for universal human values and harmonious existence. (K2) CO6: Develop as socially and ecologically responsible engineers. (K2)
IV	V23MAT07 Probability & Statistics	After Successful completion of the Course, the student will be able to: CO1: Classify the concepts of data science and its importance.(K2) CO2: Interpret the association of characteristics and through correlation and regression tools. (K3) CO3: Apply discrete and continuous probability distributions. (K2) CO4: Construct the components of a classical hypothesis test. (K3) CO5: Apply the statistical inferential methods to test the hypothesis based on small and large samplings. (K3)
IV	V23CST06 Operating Systems	After Successful completion of the Course, the student will be able to: CO1: Describe Operating System Services, System Calls, Building, Boosting and Debugging. (K2) CO2: Illustrate Process Management Concepts and CPU Scheduling Algorithms. (K3) CO3: Demonstrate Process Synchronization primitives and Deadlock handling techniques. (K3) CO4: Illustrate Memory Management Techniques and Page Replacement Algorithms. (K3) CO5: Describe File System Concepts and Mass Storage Structures. (K2)
IV	V23CST07 Database Management Systems	After Successful completion of the Course, the student will be able to: CO1: Describe Database systems, various Data models and Database architecture. (K2) CO2: Develop various real time applications using Relational algebra and Relational calculus. (K3) CO3: Illustrate Nested queries and different types of joins. (K3) CO4: Apply various Normalization techniques to refine schema. (K3) CO5: Explain Transaction management, Concurrency control and Database indexing techniques. (K2)
IV	V23CST08 Software Engineering	After Successful completion of the Course, the student will be able to: CO1: Demonstrate the evolution, development practices, and various software life cycle models. (K3) CO2: Illustrate project management techniques and explain requirements analysis and specification. (K3) CO3: Develop effective software designs, using agile methodologies and user interface principles. (K3) CO4: Apply coding practices, conduct thorough testing, and manage software quality. (K3) CO5: Distinguish the use of CASE tools, manage software maintenance, and implement software reuse strategies. (K4)
IV	V23CSL06 Operating Systems Lab	After Successful completion of the Course, the student will be able to: CO1: Practice of UNIX fundamentals, commands & system calls. (K3) CO2: Construct programs using CPU scheduling algorithms. (K3) CO3: Demonstrate IPC, semaphores, monitors, deadlocks. (K3) CO4: Develop programs to solve Page replacement algorithms, file allocation strategies and memory allocation strategies. (K3)

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IV	V23CSL07 Database Management Systems Lab	After Successful completion of the Course, the student will be able to: CO1: Construct SQL queries using DDL,DML and DCL commands. (K3) CO2: Experiment with nested queries, built-in functions and various constraints.(K3) CO3: Construct PL/SQL programs on Control structures, Cursors and Triggers. (K3) CO4: Develop a Java program to connect with a database using JDBC. (K3)
IV	V23CSSE02 Full Stack Development –I	After Successful completion of the Course, the student will be able to: CO1: Illustrate HTML elements and their attributes for designing static web pages. (K3) CO2: Apply appropriate CSS styles to HTML elements. (K3) CO3: Demonstrate JavaScript Pre-defined and User-defined Objects. (K3) CO4: Develop dynamic web pages and validate forms using JavaScript. (K3)
IV	V23CEAC01 Environmental Science	After Successful completion of the Course, the student will be able to: CO1: Illustrate multi disciplinary nature of environmental studies and various renewable and non-renewable resources. (K3) CO2: Outline flow and bio-geo- chemical cycles and ecological Pyramids. (K4) CO3: Ascertain various causes of pollution and solid waste management and related preventive measures. (K5) CO4: Discriminate Understand the rainwater harvesting, watershed management, ozone layer depletion and waste land reclamation. (K4) CO5: Illustrate the causes of population explosion, value education and welfare programs. (K3)
IV	V23ENT03 Professional Communication Skills - II	After Successful completion of the Course, the student will be able to: CO1: Demonstrate grammatical competence, analyze noun and pronoun dispositions, classify various kinds of verbs, adjectives and adverbs and identify errors in sentences; distinguish the subtle meanings of various words in different contexts, recognize similar words as well as words with contrast meanings and use them appropriately. (K3) CO2: Organize individual words into one whole sentence using new vocabulary and focus on the error analysis of prepositions and conjunctions, build conversations which befit the situations and develop pre-reading strategies to improve comprehension skills. Distinguish and acquire knowledge of using words of the same category in a sentence and learn new words that promote communicative finesse. Find errors in sentences where the modifiers are misplaced and put them at the appropriate place, use hit pair words and send an email that is concise and lucid. (K3) CO3: Recognize the easiest and best possible way of solving problems in the area of Number and Letter Series, Analogy, Classification, Coding & Decoding Symbols, Ranking and Analytical Reasoning.(K4) CO4: Investigate the different types of logics involved in Mirror and Water Images, Logical Reasoning & Arithmetic Reasoning. (K4) CO5: Find the common traps in the questions and errors likely to be made from the concepts of Blood Relations, Directions, Average, Clock and Calendar, Data Sufficiency, Permutations-Combinations and Probability.(K3)
V	V23CST09 Data Warehousing and Data Mining	After Successful completion of the Course, the student will be able to: CO1: Discuss the basic concepts and techniques of data warehousing and data mining. (K2) CO2: Demonstrate the types of the data to be mined and apply pre-processing methods on raw data. (K3) CO3: Illustrate various Classification Techniques. (K3) CO4: Discover interesting patterns, analyze supervised and unsupervised models and estimate the accuracy of the algorithms. (K3) CO5: Use different clustering techniques to cluster data. (K3)
V	V23CST10 Computer Networks	After Successful completion of the Course, the student will be able to: CO1: Illustrate fundamentals of networks, Reference Models and Transmission Media. (K2)

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		CO2: Demonstrate Error Detection & Correction and Data Link Layer protocols.(K3) CO3: Illustrate the protocols of MAC sub layer and Channelization. (K2) CO4: Utilizeconnection and connectionless services and Apply Routing and Congestion Control Algorithms. (K3) CO5: Explain Transport Layer and Application Layer protocols.(K2)
V	V23CST11 Automata and Compiler Design	After Successful completion of the Course, the student will be able to: CO1: Construct Finite Automata and Regular Expressions. (K3) CO2: Describe the Compilation Process and Lexical Analysis.(K2) CO3: Construct Top down and Bottom up Parsing Techniques. (K3) CO4: Produce Intermediate Code Generation and Runtime Environments. (K3) CO5: Explain Code Optimization and Code Generation. (K2)
V	V23CSTPE01 Object Oriented Analysis and Design	After Successful completion of the Course, the student will be able to: CO1: Illustrate the inherent complexity and attributes of software systems and Design structured solutions for complex problems by applying principles of system architecture. (K3) CO2: Use the significance and principles of object-oriented modeling and apply UML concepts to represent structural aspects of software systems using appropriate diagrams. (K3) CO3: Develop detailed class and object models and differentiate between basic and advanced structural constructs such as interfaces, packages, and roles for effective software design.(K3) CO4: Construct behavioral models such as use case, interaction, and activity diagrams to illustrate user interactions and system functionalities in real-world applications. (K3) CO5: Design comprehensive behavioral and architectural models using state charts, component, and deployment diagrams to demonstrate system dynamics and implementation strategies.(K3)
V	V23CSTPE02 Artificial Intelligence	After Successful completion of the Course, the student will be able to: CO1: Understand the fundamental concepts of Artificial Intelligence. (K2) CO2: Employ various search strategies and game-playing algorithms to solve AI problems. (K3) CO3: Apply knowledge representation techniques and probabilistic reasoning methods to model AI systems. (K3) CO4: Use logic based inference techniques and learning methods for intelligent decision- making. (K3) CO5: Demonstrate the principles of expert system architecture and knowledge acquisition in AI applications. (K3)
V	V23CSTPE03 Microprocessors & Microcontrollers	After Successful completion of the Course, the student will be able to: CO1: Understand the fundamental concepts of Artificial Intelligence. (K2) CO2: Employ various search strategies and game-playing algorithms to solve AI problems. (K3) CO3: Apply knowledge representation techniques and probabilistic reasoning methods to model AI systems. (K3) CO4: Use logic based inference techniques and learning methods for intelligent decision- making. (K3) CO5: Demonstrate the principles of expert system architecture and knowledge acquisition in AI applications. (K3)
V	V23CSTPE04 Quantum Computing	After Successful completion of the Course, the student will be able to: CO1: Understand the foundational principles of quantum mechanics relevant to computing.(K2) CO2: Explain quantum computing or related interdisciplinary fields(mathematics, physics) . (K2) CO3: Construct various physical quantum units of information. (K3) CO4: Demonstrate the various quantum Algorithms. (K3) CO5: Illustrate the role of noise in quantum systems and the necessity for error correction.(K3)

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V	V23CSL08 Data Mining Lab	After Successful completion of the Course, the student will be able to: CO1: Describe Conceptual, Logical, and Physical design of Data Warehouses OLAP applications and OLAP deployment.(K2) CO2: Develop a data warehouse or data mart to present information needed by management in a form that is usable.(K3) CO3: Practice hands-on experience working with all real data sets. (K3) CO4: Compute real data sets using popular data mining tools such as WEKA, Python Libraries. (K3) CO5: Develop ability to design various algorithms based on data mining tools. (K3)
V	V23CSL09 Computer Networks Lab	After Successful completion of the Course, the student will be able to: CO1: Develop programs on data link layer framing methods and checksum. (K3) CO2: Develop programs to implement Error detection technique and Sliding window protocol. (K3) CO3: Develop programs to implement Routing and Congestion Control Algorithms. (K3) CO4: Demonstrate various networking tools like Wireshark, NS2 and Nmap. (K3)
V	V23CSSE03 Full Stack development-2	After Successful completion of the Course, the student will be able to: CO1: Explain the key concepts of ExpressJS such as routing, HTTP methods, middleware, templating, and handling form data. (K2) CO2: Apply ExpressJS to implement cookies, sessions, authentication, database operations, and RESTful APIs in server-side applications. (K3) CO3: Develop ReactJS components using JSX, props, state, and styling to handle events and user interactions.(K3) CO4: Analyse ReactJS techniques such as conditional rendering, forms, routing, hooks, and screen updates to manage dynamic content. (K4) CO5: Apply MongoDB CRUD operations to create and manage databases, collections, and documents for full-stack applications. (K3)
V	V23CSES01 Master Coding and Competitive Programming - Part-1	After Successful completion of the Course, the student will be able to: CO1: Apply Mathematical reasoning and number theory to solve real world problems in linear time. (K3) CO2: Use of modular arithmetic, to solve complex problems in linear time, logarithmic. (K3) CO3: Use of Prime Factorization and complex solve problems. (K3) CO4: Analyse different techniques including sieve to find prime numbers and evaluate efficiency of these methods. (K4) CO5: Experiment with Hashing and searching techniques to solve problems on Arrays in Linear time. (K3)
V	V23CSES02 User Interface Design using Flutter	After Successful completion of the Course, the student will be able to: CO 1: Apply Dart and Flutter setup procedures to build basic mobile application components. (K3) CO 2: Develop responsive user interfaces using layout widgets and media queries.(K3) CO 3: Apply navigation techniques and state management strategies to enhance functionality in Flutter applications. (K3) CO 4: Construct functional mobile app components using API integration, animations, custom widgets, and debugging tools. (K3)
V	V23ENT04 English for Employability	After Successful completion of the Course, the student will be able to: CO1: Demonstrate proficiency in verbal ability, including word meanings, synonyms, antonyms, and contextual usage, to enhance their communication skills and excel in competitive exams. (K3) CO2: Effectively apply various reading techniques and comprehension skills, including literal, inferential, critical, evaluative, and appreciative reading, to accurately understand and interpret complex texts. (K3) CO3: Apply knowledge and skills to participate effectively in group discussions on various topics, showcasing critical thinking, communication, and teamwork abilities. (K3) CO4: Construct a well-organized and compelling resume tailored to specific job descriptions and industries. (K3) CO5: Analyse the skills required to excel in an interview and identify areas for improvement. (K1)

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VI	V23CST12 Machine Learning	After Successful completion of the Course, the student will be able to: CO1: Explain machine learning paradigms, data preparation, and model evaluation. (K2) CO2: Apply distance-based techniques and algorithms to classify and regress data using proximity measures. (K3) CO3: Construct decision tree and Bayesian models to perform classification and regression. (K3) CO4: Demonstrate linear discriminants, SVMs, and neural networks to solve classification and regression problems in machine learning. (K3) CO5: Apply ensemble techniques and graphical model-based approaches to develop predictive and sequential models for real-world problems. (K3)
VI	V23CST13 Cloud Computing	After Successful completion of the Course, the student will be able to: CO 1: Explain the evolving utility computing model called cloud computing. (K2) CO 2: Explain the different service models of cloud computing, such as IaaS, PaaS, and SaaS. (K2) CO 3: Explain the core concepts of cloud technologies like distributed computing, service-oriented architecture, and virtualization. (K2) CO 4: Analyze the security and other challenges in cloud computing. (K4) CO 5: Apply advanced concepts such as containers, serverless computing, and cloud-centric Internet of Things. (K3)
VI	V23CST14 Cryptography & Network Security	After Successful completion of the Course, the student will be able to: CO1: Explain fundamentals of security concepts and mathematical support of Cryptography. (K2) CO2: Discuss Symmetric cryptosystems. (K2) CO3: Discuss Asymmetric cryptosystems. (K2) CO4: Apply Hash functions & Digital Signatures to provide authentication and integrity. (K3) CO5: Demonstrate various methods of Mutual trust and mail security Network Security Scenarios. (K3)
VI	V23CSTPE05 Software Testing Methodologies	After Successful completion of the Course, the student will be able to: CO1: Describe Path Testing using flow graphs. (K2) CO2: Demonstrate Transaction Flow, Dataflow and Domain Testing Techniques. (K3) CO3: Apply Path Expressions and Logic-Based Testing. (K3) CO4: Demonstrate State Transition Testing Using State Graphs. (K3) CO5: Implement Graph Matrices and Testing Tools for Test Case Generation, Automation. (K3)
VI	V23CSTPE06 Cyber Security	After Successful completion of the Course, the student will be able to: CO1: Explain the fundamental concepts of cybercrime, its origins, types, and the challenges posed by modern devices and technologies. (K2) CO2: Demonstrate various tools and techniques used in committing cybercrimes, including phishing, malware, spoofing, session hijacking, and social engineering. (K2) CO3: Demonstrate knowledge of cybercrime investigation processes including digital evidence collection, eDiscovery, email/IP tracking, and the use of encryption / decryption tools. (K2) CO4: Apply computer forensic techniques and use current tools to investigate and analyze digital evidence on different platforms such as Windows, Linux, and mobile devices. (K3) CO5: Demonstrate the legal framework related to cybercrime, with a focus on the Indian IT Act, its challenges, amendments, and the global legal scenario. (K2)
VI	V23CSTPE07 DevOps	After Successful completion of the Course, the student will be able to: CO1: Explain the architecture, principles, and lifecycle of DevOps and how it integrates with Agile practices and automation tools.. (K2) CO2: Apply GIT commands and workflows to manage source code collaboratively and

Semester	Course Code & Course Name	Course Outcomes
		analyze code quality using SonarQube. (K3) CO3: Develop and manage continuous integration pipelines using Jenkins to automate build and testing processes. (K3) CO4: Implement containerized applications using Docker and integrate testing tools for continuous delivery. (K3) CO5: Use Ansible and OpenShift to automate application deployment and manage container orchestration in CI/CD pipelines. (K3)
VI	V23CSTPE08 Mobile Adhoc Networks	After Successful completion of the Course, the student will be able to: CO1: Explain MANETs, their uses, challenges, and types of MAC protocols. (K2) CO2: Apply suitable routing and transport protocols in ad hoc wireless networks. (K3) CO3: Describe common security threats and basic protection methods in MANETs. (K2) CO4: Use proper methods to collect and send data in wireless sensor networks. (K3) CO5: Operate tools and software to build secure wireless sensor network systems. (K3)
VI	V23CSTPE09 Software Project Management	After Successful completion of the Course, the student will be able to: CO1: Explain conventional and agile software methods with their effect on software economics. (K2) CO2: Explain modern software practices and identify life cycle phases and artifacts. (K2) CO3: Describe software architectures, workflows, and checkpoints in iterative development. (K2) CO4: Illustrate planning techniques and identify project structures and automation elements. (K3) CO5: Describe software metrics and capability models to improve quality. (K2)
VI	V23CSTPE10 Natural Language Processing	After Successful completion of the Course, the student will be able to: CO1: Explain the fundamental concepts of language modeling, English morphology, and spelling correction techniques using regular expressions and finite-state automata. (K2) CO2: Apply N-gram models and part-of-speech tagging techniques using stochastic and rule-based approaches to perform word-level language analysis. (K3) CO3: Construct syntactic parsers using context-free grammars and apply probabilistic parsing methods such as PCFG and CYK for syntactic analysis. (K3) CO4: Describe semantic representation techniques and methods for word sense disambiguation using supervise and knowledge-based approaches. (K2) CO5: Apply discourse segmentation and reference resolution techniques using lexical resources like WordNet, Penn Treebank, and Brill's Tagger for effective text analysis. (K3)
VI	V23CSTPE11 Big Data Analytics	After Successful completion of the Course, the student will be able to: CO1: Describe big data concepts, applications, and related technologies. (K2) CO2: Use NoSQL models and perform basic operations in Cassandra. (K3) CO3: Develop data processing tasks using Hadoop and Hive. (K3) CO4: Build data analytics workflows using Apache Spark. (K3) CO5: Implement streaming and tuning techniques in Spark. (K3)
VI	V23CSTPE12 Distributed Operating System	After Successful completion of the Course, the student will be able to: CO1: Describe the basic concepts of distributed operating systems. (K2) CO2: Understand the Communication using RPC. (K2) CO3: Demonstrate the concept of shared memory. (K3) CO4: Explain resource management techniques. (K2) CO5: Illustrate Distributed file system and fault tolerance performance. (K2)
VI	V23CSL10 Cloud Computing Lab	After Successful completion of the Course, the student will be able to: CO1: Demonstrate various service types, delivery models and technologies of a cloud computing environment. (K3) CO2: Classify cloud services based on virtual machines and containers in the cloud offerings. (K2) CO3: Identify and outline key challenges associated with a cloud-based application. (K2) CO4: Illustrate advanced cloud concepts such as server less computing and cloud simulation. (K3) CO5: Apply suitable programming paradigms to solve real world and scientific

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		problems using cloud services. (K3)
VI	V23CSL11 Machine Learning Lab using Python	After successful completion of the course, the student will be able to: CO1: Identify various Python libraries used in Machine Learning.(K2) CO2: Construct non-probabilistic classifiers using Python Programming.(K3) CO3: Implement probabilistic classifiers using Python Programming.(K3) CO4: Apply ensemble methods for classification tasks. (K3) CO5: Demonstrate neural and sequential models using Python.(K3)
VI	V23CSSE04 Master Coding and Competitive Programming - Part-2	After Successful completion of the Course, the student will be able to: CO1: Apply Divide and Conquer algorithm technique to solve complex in logarithmic time. (K3) CO2: Apply Greedy method to solve Optimization and decision making problems. (K3) CO3: Apply Backtracking Algorithm technique to find combinatorial problems. (K4) CO4: Experiment with Dynamic Programming Algorithm technique to solve Problems that uses Optimal substructures. (K3) CO5: Develop programs using Linked List Graphs, DFS and BFS techniques. (K3)
VI	V23ENT05 Technical Paper Writing &IPR	After Successful completion of the Course, the student will be able to: CO1: Understand the structure of the technical paper and its components. (K2) CO2: Review the literature and acquire the skills to write a technical paper for first submission. (K2) CO3: Understand the process and development of IPR. (K2) CO4: Create awareness about the scope of patent rights. (K6) CO5: Analyze the new developments in IPR include latest software. (K4)